

Haowen Lai

hwlai@seas.upenn.edu | [Homepage](#) | [LinkedIn](#) | [Google Scholar](#)

University of Pennsylvania, Philadelphia, PA, 19104, U.S.A

EDUCATION

University of Pennsylvania

Ph.D. in Computer and Information Science

Philadelphia, PA, USA

Sept. 2022 - Expected May 2027

Tsinghua University

M.S. in Automation

Beijing, China

Sept. 2019 - Jun. 2022

Tongji University

B.E. in Automation (GPA: 4.85/5.00, **ranking: 1/79**)

Shanghai, China

Sept. 2015 - Jul. 2019

RESEARCH INTERESTS

My research focuses on building robust **multimodal wireless sensing and perception systems**. I develop **RF and radar perception**, including **foundation models for RF sensing**, that integrate wireless signals with vision and other sensing modalities to understand **geometry, semantics, materials, and human activities** in challenging real-world environments. I am also interested in applying RF sensing to **health monitoring, smart homes, and human-centered computing**, where privacy, robustness, and continuous operation are critical. My work combines **wireless systems, physics-guided signal processing, learning-based imaging, and multimodal learning**. More broadly, my interests also include **LLM, SLAM, localization, 3D reconstruction, imaging, segmentation, and detection**.

PUBLICATIONS

- [1] **H. Lai**, Z. Lan, D. Hu, and M. Zhao, "Surface Characterization with mmWave Signals," *The 24th ACM International Conference on Mobile Systems, Applications, and Services (MobiSys)*. 2026. [\[pdf\]](#)
- [2] Z. Lan, Y. Tang, Y. Wang, **H. Lai**, Y. Hao, M. Zhao, "Building Audio-Visual Digital Twins with Smartphones," *The 24th ACM International Conference on Mobile Systems, Applications, and Services (MobiSys)*. 2026. [\[pdf\]](#)
- [3] **H. Lai**, Z. Lan, and M. Zhao, "Non-Line-of-Sight 3D Reconstruction with Radar," *The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS)*. 2025. [\[pdf\]](#)
- [4] **H. Lai**, Z. Zheng, and M. Zhao, "RF-Based 3D SLAM Rivaling Vision Approaches," *The 31st Annual International Conference on Mobile Computing and Networking (MobiCom)*. 2025. 🏆 **Best Artifact Award**. [\[pdf\]](#)
- [5] **H. Lai**, G. Luo, Y. Liu and M. Zhao, "Enabling Visual Recognition at Radio Frequency," *The 30th Annual International Conference on Mobile Computing and Networking (MobiCom)*. 2024. 🏆 **Best Demo Award** 🏆 **First Place in SRC** 🏆 **Second Place in ACM SRC Grand Finals**. [\[pdf\]](#)
- [6] R. Yan, R. Deng, **H. Lai**, W. Zhang, Z. Shi and Y. Zhong, "Homicidal Chauffeur Reach-Avoid Games via Guaranteed Winning Strategies," *IEEE Transactions on Automatic Control (TAC)*. 2023. [\[pdf\]](#)
- [7] P. Yin, S. Zhao, **H. Lai**, R. Ge, J. Zhang, H. Choset and S. Scherer, "Automerge: A framework for map assembling and smoothing in city-scale environments," *IEEE Transactions on Robotics (T-RO)*. 2023. [\[pdf\]](#)
- [8] **H. Lai**, P. Yin and S. Scherer, "AdaFusion: Visual-LiDAR Fusion with Adaptive Weights for Place Recognition," *IEEE Transactions on Robotics (RA-L)*. 2022. [\[pdf\]](#)

- [9] W. Liang, **H. Lai**, Z. Shi and Y. Zhong, "Global Registration of Point Cloud Maps with Low-overlap Regions," *IEEE Chinese Control Conference (CCC)*. 2022. [\[pdf\]](#)
- [10] **H. Lai**, R. Yan, W. Zhang, Z. Shi and Y. Zhong, "Reach-Avoid Differential Games via Finite-Time Heading Tracking," *IEEE Conference on Decision and Control (CDC)*, 2021. [\[pdf\]](#)
- [11] **H. Lai**, W. Liang, R. Yan, Z. Shi and Y. Zhong, "LiDAR-Inertial based Localization and Perception for Indoor Pursuit-Evasion Differential Games," *IEEE Chinese Control Conference (CCC)*, 2021. [\[pdf\]](#)
- [12] **H. Lai**, Q. Kang, L. Pan et al., "A Novel Scale Recognition Method for Pointer Meters Adapted to Different Types and Shapes," *IEEE International Conference on Automation Science and Engineering (CASE)*, 2019. [\[pdf\]](#)

PATENTS

- [1] M. Zhao, G. Luo, Y. Liu and **H. Lai**, "Methods, Systems, and Computer Readable Media for Providing 3D Imaging Using Radio Frequencies," US Patent Application 63/626,860, filed Jan. 2024. Patent Pending.
- [2] B. Wen, J. Zhan, S. Liang, T. Lu, Q. Xiong, X. Jiang and **H. Lai**, "Registration Method based on CNN Point Cloud Object Detection," China Patent CN112700479B, authorized Feb. 2024.
- [3] **H. Lai** and X. Jiang, "A Map Construction Method and a LiDAR-inertial Odometry," China Patent CN113358112B, authorized Jan. 2023.
- [4] X. Jiang, Q. Xiong, S. Liang, J. Zhang, **H. Lai** and B. Wen, "Method and System for Localization based on Sensor Fusion," China Patent CN113375666B, authorized Dec. 2022.
- [5] **H. Lai**, J. Jiang, J. Chen and L. Jiang, "Method and System for Camera Calibration based on Deep Learning," China Patent CN109493389B, authorized Nov. 2021.
- [6] **H. Lai**, J. Chen, J. Jiang and L. Jiang "Method and System for Robot Arm Controlling based on Deep Learning," China Patent CN109352649B, authorized Jul. 2021.
- [7] Q. Kang and **H. Lai**, "Automatic Reading Method for Pointer Meters based on Scale Seeking," China Patent CN109993166B, authorized Oct. 2020.

HONORS AND AWARDS

- | | |
|---|------------------|
| • Best Artifact Award in ACM MobiCom | 2025 |
| • Second Place in ACM SRC Grand Finals (2/334, 21 ACM conferences) | 2025 |
| • First Place in ACM MobiCom Student Research Competition (SRC) | 2024 |
| • Best Demo Award in ACM MobiCom | 2024 |
| • Outstanding Undergraduate of Shanghai, (top 5%) | 2019 |
| • Excellent Student of Tongji University, (top 5%) | 2018, 2017& 2016 |
| • Phoenix Contact Scholarship, (excellence in study and research) | 2018 |
| • Siemens Scholarship, (top 5%) | 2018 |
| • The National First Prize in Intel Cup ESDC, (top 8% of 164 teams) | 2018 |
| • The Third Prize of the 16th Tan Kah-Kee Youth Invention Award | 2018 |
| • Qidi Scholarship of Tongji University, (excellence in research) | 2017 |
| • National Scholarship for Undergraduate, (top 2.5%) | 2017 |
| • The First Prize of Tongji Scholarship of Excellence, (top 5%) | 2016 |

MEDIA COVERAGE

- Penn Today, [Robots that can see around corners using radio signals and AI](#)
- BBC, [Scientists work on 'superhuman' vision systems for robots](#)
- Penn Today, [Giving robots superhuman vision using radio signals](#)
- MIT Technology Review, [Scientists are using large AI models for sensor design, creating a high-resolution radio frequency imaging system that achieves high precision imaging in dynamic environments](#)

RESEARCH EXPERIENCES

PhD Student, advisor: Prof. Mingmin Zhao **University of Pennsylvania – WAVES Lab**

Surface Characterization with mmWave Signals

Mar. 2025 – Dec. 2025

- Built the first fully automatic RF-based surface characterization system for in-the-wild scenarios, capable of both identification and property estimation.
- Showed that coherent imaging serves as a new representation for RF material sensing
- Derived a physics-grounded imaging model linking surface properties to observed profiles
- Proposed a voting strategy that resolves mixed-material ambiguities and produces consistent material maps

Non-Line-of-Sight 3D Reconstruction with Radar

Dec. 2024 – May. 2025

- Built the first practical and mobile system for NLOS 3D scene reconstruction.
- Proposed a novel two-stage pipeline that decouples signal interpretation from spatial reasoning.
- Developed a physics-inspired architecture that models ray tracing and surface normal to reverse multi-bounce mirroring effects for NLOS reconstruction.

RF-based 3D SLAM Rivaling Vision Approaches

Nov. 2023 – Sept. 2024

- Built an RF-based 3D SLAM system that achieves centimeter-level localization and mapping with high-fidelity details comparable to vision-based approaches
- Introduced a novel and non-intrusive uncertainty estimation method for ML-based RF sensing, offering out-of-the-box usage and mitigating the influence of output errors
- Proposed an implicit occupancy field optimized with uncertainty-aware maximum likelihood estimation (MLE), effectively handling noise and remaining efficient

Enabling Visual Recognition at Radio Frequency

Dec. 2022 – Aug. 2023

- Developed a novel RF imaging system that brings RF resolution close to that of LiDAR, while providing resilience against conditions challenging for optical signals
- Presented a novel design that integrates a COTS mmWave sensor with a motor to significantly enhance sensing resolution and FoV, while ensuring that the system remains compact, low-cost, and practical for mobile robots
- Proposed a novel robot motion estimation algorithm that accurately estimates and compensates for robot motion, allowing for coherent combination of radio signals
- Introduced a learning model that effectively enhances vertical resolution using high azimuth and range resolutions, while maintaining efficiency with 2D convolutions

Research Intern, advisor: Prof. Sebastian Scherer

Carnegie Mellon University – Air Lab

AutoMerge: Automatic Multi-agent Cooperated Map Merging System

Dec. 2021 – Apr. 2022

- Researched on the problem of merging large-scale and low-overlap 3D point cloud maps without any prior information about their initial relative poses
- Developed an offline version, including a global localization frontend resistant to viewpoint changes and a pose

graph optimization backend for fusing all local maps

Adaptive Visual-LiDAR Fusion for Place Recognition and Global Localization

May. 2021 - Oct. 2021

- Researched on the problem of place recognition with the method that adaptively selects and weights LiDAR and visual features according to different environments
- Proposed a weight generation branch including multi-scale attention and two-stage fusion that could produce adaptive weights to adjust the importance and contribution of features of different modalities
- Reproduced the feature extraction backbone and improved it by utilizing batch normalization (BN)
- Built the whole network including data preparation, network training and evaluation in Python and PyTorch

Research Assistant, advisor: Prof. Yisheng Zhong

Tsinghua University – UAV Lab

LiDAR-Inertial based Self-Localization and Target Perception for Robots

Jan. 2021 - Jul. 2021

- Utilized a tightly-coupled LiDAR inertial odometry for self-localization in unknown GPS-denied environment
- Proposed a model fitting based target perception method to determine the position of opponents from point clouds
- Applied autonomous localization and perception to indoor multi-robot pursuit-evasion games

Intelligent and Unmanned Indoor Navigation Robot

Jan. 2020 - Sept. 2020

- Tested and compared the results of mapping, localization and loop closure detection of several SLAM methods such as gmapping, RGBD SLAM v2, RTAB-MAP, etc.
- Built the wheel robots and developed a multi-robot system including visual SLAM and navigation based on ROS
- Researched on the problems of robot navigation that avoids obstacles of lower objects via RGB-D camera, and autonomous exploration with boundary search based on BFS

Research Assistant, advisor: Prof. Qi Kang

Tongji University – BitAI Lab

Vision-Based Automatic Reading System for Pointer Meters

Nov. 2018 - Aug. 2019

- Trained the SSD + MobileNet object detection network on the MS COCO dataset to locate the dial
- Proposed an Adaptive Scale Seeking algorithm based on image characteristics to automatically read both uniform and non-uniform scale meters in constraint environment
- Built an image collection device with Wi-Fi connection based on STM32 Microcontroller Unit (MCU)
- Developed a user-friendly data management GUI for monitoring, recording and controlling the whole system

INDUSTRIAL EXPERIENCE

Google, Research Intern

Mountain View, CA, USA

Health and Home (H2O),

May 2026 - Aug. 2026

- Researched on radar foundation models and multimodal sensing for health applications
- Developed multimodal data collection systems and applications, initial prototypes, and training pipelines
- Presented demonstration of health sensing and monitoring on targeted devices

PROJECTS

Team Leader

Intel Cup ESDC 2018

Self-Learning Model-Free Robot Arm System for Grabbing and Classification

Jan. 2018 - Aug. 2018

- Won the national first prize (top 8% of 164 teams) of the *Intel Cup ESDC 2018* competition
- Responsible for the visual localization of robot joints and target cubes based on ArUco markers and image

segmentation, respectively

- Developed the self-learning control model as a multi-layer fully connected neural network with TensorFlow, and collected motor control values and the corresponding target coordinates for training
- Implemented the communication between the two manipulators via Socket and that between the host computer and the control board through CAN bus

Team Leader, advisor: Prof. Fanhuai Shi

National Innovation Project

Follow You up: Selfie UAV with Gesture Interaction

Apr. 2017 - Apr. 2018

- Won the third prize of the *16th Tan Kah-Kee Youth Invention Award (Shanghai)*
- Enable real-time detection and tracking of human body using RGB-D camera and machine learning, and control the drone to follow human based on the ROS framework
- Realized the segmentation of gesture based on skin color, and used TensorFlow to build CNN for recognition

TEACHING EXPERIENCES

Teaching Assistant

University of Pennsylvania

- CIS 3990: Mobile and IoT Computing

Spring 2024

Teaching Assistant

Tsinghua University

- 20250193-2: Operations Research
- 20250193-3: Operations Research
- 10421055-10: Calculus A (1)

Spring 2021

Spring 2020

Fall 2019

PROFESSIONAL ACTIVITIES

Paper Review

- Annual Conference on Neural Information Processing Systems (NeurIPS)
- The International Journal of Robotics Research (IJRR)
- IEEE International Conference on Robotics & Automation (ICRA)
- IEEE Transactions on Mobile Computing (TMC)
- IEEE Transactions on Automation Science and Engineering (T-ASE)
- IEEE Transactions on Control Systems Technology (TCST)
- IEEE Robotics and Automation Letters (RA-L)
- Elsevier Measurement

Organizing Committee

- Co-organizer of ICRA 2022 General Place Recognition Competition

Feb. 2022 - Apr. 2022

Presentations, Posters & Demos

- Real-time Demonstration at Google H&H Intern Showcase
"Radar Foundation Models"
Mountain View, CA, USA, Aug. 2026
- Video Recording Presentation at the ACM MobiSys
"Surface Characterization with mmWave Signals"
Cambridge, UK, Jun. 2026
- Poster Presentation at the NeurIPS
"Non-Line-of-Sight 3D Reconstruction with Radar"
San Diego, CA, USA, Dec. 2025

- Oral Presentation & Demo at the ACM MobiCom Washington, D.C., USA, Nov. 2024
“Enabling Visual Recognition at Radio Frequency”
- Remote Presentation at the IEEE Conference on Decision and Control Austin, TX, USA, Dec. 2021
“Reach-Avoid Differential Games via Finite-Time Heading Tracking”
- Oral Presentation at the IEEE Chinese Control Conference Shanghai, China, Jul. 2021
“LiDAR-Inertial based Localization and Perception for Indoor Pursuit-Evasion Differential Games”
- Oral Presentation at the IEEE CASE Vancouver, Canada, Aug. 2019
“A Novel Scale Recognition Method for Pointer Meters Adapted to Different Types and Shapes”

SKILLS

Language: English – Fluent, Chinese – Native, Cantonese – Native

Programming: Python, C, C++, MATLAB

Library: PyTorch, Tensorflow, Jax, ROS, OpenCV, PCL, Open3D, Eigen, GTSAM, Qt

Technical: laser cutting (with DraftSight), 3D printing (with SolidWorks), Docker, video editing